



INSTRUMENTACIÓN DE CONTROLADORES

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SEGG-U2-P3G-2 Instrumentación de Controladores

Objetivo

Instrumentar todos los controladores utilizando ILogger<T> siguiendo buenas prácticas de desarrollo seguro.

Ejemplo de Instrumentación

```
private readonly ILogger<SearchController> _logger;
_logger.LogInformation("Inicio Search.Index");
_logger.LogInformation("Usuario:{User}
IP:{IP}",user,HttpContext.Connection.RemoteIpAddress);
try{
    // lógica
}
catch(Exception ex){
    _logger.LogError(ex,"Error");
}
_logger.LogInformation("Fin Search.Index");
```

Cada acción deberá registrar

- Entrada.
- Salida.
- Tiempo de ejecución.
- Usuario.
- IP.
- Parámetros (excepto contraseñas).
- Warnings.
- Errores.
- Eventos de autenticación.
- Excepciones.

Controladores a instrumentar

- HomeController

```

41     }
42     public IActionResult Privacy()
43     {
44         var stopwatch = Stopwatch.StartNew();
45         var ip = HttpContext.Connection.RemoteIpAddress?.ToString() ?? "Desconocida"; "Desconocida": Unknown word.
46         var usuario = HttpContext.Session.GetString("User") ?? "Anónimo"; "usuario": Unknown word.
47
48         _logger.LogInformation("Inicio HomeController.Privacy | Usuario: {Usuario} | IP: {IP}", usuario, ip); "Inicio": Unknown word.
49
50         try
51         {
52             return View();
53         }
54         catch (Exception ex)
55         {
56             _logger.LogError(ex, "Error en HomeController.Privacy");
57             throw;
58         }
59         finally
60         {
61             stopwatch.Stop();
62             _logger.LogInformation("Fin HomeController.Privacy | Tiempo de ejecución: {TiempoMs} ms", stopwatch.ElapsedMilliseconds); "Tiempo": Unknown word.
63         }
64     }

```

- SearchController

```

25     public IActionResult Index(string search)
26     {
27         var stopwatch = Stopwatch.StartNew();
28         var ip = HttpContext.Connection.RemoteIpAddress?.ToString() ?? "Desconocida"; "Desconocida": Unknown word.
29         var usuario = HttpContext.Session.GetString("User") ?? "Anónimo"; "usuario": Unknown word.
30
31         _logger.LogInformation("Inicio SearchController.Index | Usuario: {Usuario} | IP: {IP} | Término Búsqueda: {Search}", usuario, ip, search); "Inicio": Unknown word.
32
33         try
34         {
35             if (string.IsNullOrEmpty(search))
36             {
37                 return View(new List<User>());
38             }
39
40             // Genera un Warning si alguien busca cadenas muy largas que puedan ser intentos de inyección "alguien": Unknown word.
41             if (search.Length > 50)
42             {
43                 _logger.LogWarning("Búsqueda sospechosamente larga por el usuario {Usuario} desde la IP {IP}", usuario, ip); "Búsqueda": Unknown word.
44             }
45
46             var users = _db.Users
47                 .Where(u => u.Username.Contains(search))
48                 .ToList();
49
50             return View(users);
51         }
52         catch (Exception ex)
53         {
54             _logger.LogError(ex, "Error en SearchController.Index buscando '{Search}' por usuario {Usuario}", search, usuario); "buscando": Unknown word.
55             throw;
56         }
57         finally
58         {
59             stopwatch.Stop();
60             _logger.LogInformation("Fin SearchController.Index | Tiempo de ejecución: {TiempoMs} ms", stopwatch.ElapsedMilliseconds); "Tiempo": Unknown word.
61         }
62     }

```

- AuthController

```

[HttpPost]
public IActionResult Login(string username, string password)
{
    var stopwatch = Stopwatch.StartNew();
    var ip = HttpContext.Connection.RemoteIpAddress?.ToString() ?? "Desconocida"; "Desconocida": Unknown word.

    // Log de entrada OMITIENDO la contraseña "entrada": Unknown word.
    _logger.LogInformation("Inicio AuthController.Login (POST) | Intento de usuario: {Username} | IP: {IP} | Password: [REDACTED]", username, ip); "Inicio": Unknown word.

    try
    {
        var user = _db.Users.FirstOrDefault(u => u.Username == username);

        if (user == null || !BCrypt.Net.BCrypt.Verify(password, user.PasswordHash))
        {
            _logger.LogWarning("EVENTO AUTH: Credenciales inválidas para el intento de acceso del usuario '{Username}' desde la IP {IP}", username, ip); "EVENTO": Unknown word.
            ViewBag.Error = "Credenciales inválidas"; "Credenciales": Unknown word.
            return View();
        }

        _logger.LogInformation("EVENTO AUTH: Inicio de sesión exitoso para el usuario '{Username}' desde la IP {IP}", username, ip); "EVENTO": Unknown word.

        HttpContext.Session.SetString("User", user.Username);
        HttpContext.Session.SetInt32("UserId", user.Id);
        return RedirectToAction("Dashboard");
    }
    catch (Exception ex)
    {
    }
}

```

• ApiController

```

[HttpGet("user/{id}")]
public IActionResult GetUser(int id)
{
    var stopwatch = Stopwatch.StartNew();
    var ip = HttpContext.Connection.RemoteIpAddress?.ToString() ?? "Desconocida"; "Desconocida": Unknown word.
    var usuario = HttpContext.Session.GetString("User") ?? "Anónimo"; "usuario": Unknown word.

    _logger.LogInformation("Inicio ApiController.GetUser | Usuario: {Usuario} | IP: {IP} | Param id: {Id}", usuario, ip, id); "Inicio": Unknown word.

    try
    {
        var currentUserId = HttpContext.Session.GetInt32("UserId");
        if (!currentUserId.HasValue)
        {
            _logger.LogWarning("Intento de acceso no autorizado a la API GetUser desde IP: {IP}", ip); "Intento": Unknown word.
            return Unauthorized();
        }

        if (id != currentUserId.Value)
        {
            _logger.LogWarning("El usuario {Usuario} intentó acceder a los datos de otro id ({Id})", usuario, id); "usuario": Unknown word.
            return Forbid();
        }

        var user = _db.Users.Find(id);
        if (user == null) return NotFound();

        return Ok(new { user.Id, user.Username, user.Email });
    }
    catch (Exception ex)
    {
        _logger.LogError(ex, "Error en ApiController.GetUser para el usuario {Usuario}", usuario); "usuario": Unknown word.
        return StatusCode(500, "Error interno del servidor"); "interno": Unknown word.
    }
    finally
    {
        stopwatch.Stop();
        _logger.LogInformation("Fin ApiController.GetUser | Tiempo de ejecución: {TiempoMs} ms", stopwatch.ElapsedMilliseconds); "Tiempo": Unknown word.
    }
}

```

Pruebas

HomeController

• Generar Error.

02 Jul 2026 22:18:50.432

02 Jul 2026 22:18:50.429

02 Jul 2026 22:18:50.425

02 Jul 2026 22:18:50.422

02 Jul 2026 22:18:50.413

●

An unhandled exception has occurred while executing the request.

Executed endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'

Executed action VulnerableApp.Controllers.HomeController.Index (VulnerableApp) in 30.1596ms

Fin HomeController.Index | Tiempo de ejecución: 16 ms

●

Error en HomeController.Index

Event

Level (Error)

Type (0x9A16A410)

Trace (ff5c...)

Export

✓

✕

ActionId

404c8155-39da-4f40-a987-30ce47efb09c

✓

✕

ActionName

VulnerableApp.Controllers.HomeController.Index (VulnerableApp)

✓

✕

ConnectionId

0HNMON8ATFKG3

✓

✕

MachineName

DESKTOP-H4CTBTI

✓

✕

RequestId

0HNMON8ATFKG3:00000003

✓

✕

RequestPath

/

✓

✕

SourceContext

VulnerableApp.Controllers.HomeController

System.Exception: Error provocado para evidencia en HomeController

at VulnerableApp.Controllers.HomeController.Index() in

C:\Users\valec\OneDrive\Documents\git\VulnerableApp\Controllers\HomeController.cs:line 28

at lambda_method5(Closure, Object, Object[])

at

Microsoft.AspNetCore.Mvc.Infrastructure.ActionMethodExecutor.SyncActionResultExecutor.Execute(ActionContext

actionContext, IActionResultTypeMapper mapper, ObjectMethodExecutor executor, Object controller, Object[]

arguments)

at Microsoft.AspNetCore.Mvc.Infrastructure.ActionMethodExecutor.SyncActionResultExecutor.Execute(ActionContext

SearchController

• Ejecutar todas las acciones de cada controlador.

02 Jul 2026 22:33:42.131

02 Jul 2026 22:33:42.124

●

Connection id "0HNMONCBKIK4D" completed keep alive response.

Executed endpoint 'VulnerableApp.Controllers.SearchController.Index (VulnerableApp)'

Event

Level (Information)

Type (0xE6E435BF)

Trace (005c...)

Export

✓

✕

ConnectionId

0HNMONCBKIK4D

✓

✕

EndpointName

VulnerableApp.Controllers.SearchController.Index (VulnerableApp)

✓

✕

EventId.Id

1

✓

✕

EventId.Name

ExecutedEndpoint

✓

✕

MachineName

DESKTOP-H4CTBTI

✓

✕

RequestId

0HNMONCBKIK4D:00000005

✓

✕

RequestPath

/search

✓

✕

SourceContext

Microsoft.AspNetCore.Routing.EndpointMiddleware

02 Jul 2026 22:33:42.116

02 Jul 2026 22:33:42.113

Executed action VulnerableApp.Controllers.SearchController.Index (VulnerableApp) in 133.7221ms

Executed ViewResult - view Index executed in 52.4182ms.

• Generar Warning

02 Jul 2026 22:32:51.321

●

Búsqueda sospechosamente larga por el usuario user1 desde la IP ::1

Event

Level (Warning)

Type (0xB6606E48)

Trace (df26...)

Export

✓

✕

ActionId

0f4bcd2-f8ce-4786-a2b0-f71e6030764c

✓

✕

ActionName

VulnerableApp.Controllers.SearchController.Index (VulnerableApp)

✓

✕

ConnectionId

0HNMONCBKIK4G

✓

✕

IP

::1

✓

✕

MachineName

DESKTOP-H4CTBTI

✓

✕

RequestId

0HNMONCBKIK4G:00000002

✓

✕

RequestPath

/search

✓

✕

SourceContext

VulnerableApp.Controllers.SearchController

✓

✕

Usuario

user1

• Generar Error.

02 Jul 2026 22:21:20.450

● Error en SearchController.Index buscando 'null' por usuario Anónimo

Event Level (Error) Type (0x152372C4) Trace (e2ae...) Export

✓ X	ActionId	26c7228e-ba9b-4a85-8baa-88eb2df8d6fc
✓ X	ActionName	VulnerableApp.Controllers.SearchController.Index (VulnerableApp)
✓ X	ConnectionId	0HNMON8ATFKG5
✓ X	MachineName	DESKTOP-H4CTBTI
✓ X	RequestId	0HNMON8ATFKG5:00000001
✓ X	RequestPath	/search
✓ X	Search	
✓ X	SourceContext	VulnerableApp.Controllers.SearchController
✓ X	Usuario	Anónimo

System.Exception: Error provocado para evidencia en SearchController
at VulnerableApp.Controllers.SearchController.Index(String search) in
C:\Users\vaiec\OneDrive\Documentos\git\VulnerableApp\Controllers\SearchController.cs:line 35
at lambda_method7(Closure, Object, Object[])
at

AuthController

- Ejecutar todas las acciones de cada controlador.

Validar Contraseña y Credenciales erróneas

02 Jul 2026 22:30:34.392

02 Jul 2026 22:30:34.389

02 Jul 2026 22:30:34.384

02 Jul 2026 22:30:34.381

02 Jul 2026 22:30:34.378

02 Jul 2026 22:30:34.377

02 Jul 2026 22:30:34.369

02 Jul 2026 22:30:34.365

- Created DbCommand for 'ExecuteReader' (3ms).
- Creating DbCommand for 'ExecuteReader'.
- Opened connection to database 'VulnerableDb' on server '(localdb)\mssqllocaldb'.
- Opening connection to database 'VulnerableDb' on server '(localdb)\mssqllocaldb'.
- Created DbConnection. (1ms).
- Creating DbConnection.
- Entity Framework Core 10.0.8 initialized 'AppDbContext' using provider 'Microsoft.EntityFrameworkCore.SqlServer:10.0...

Inicio AuthController.Login (POST) | Intento de usuario: user1 | IP: ::1 | Password: [REDACTED]

02 Jul 2026 22:30:34.689

● EVENTO AUTH: Credenciales inválidas para el intento de acceso del usuario 'user1' desde la IP ::1

Event Level (Warning) Type (0x96205177) Trace (15d8...) Export

✓ X	ActionId	f5b029f9-273d-4031-960b-ae0ff21d0995
✓ X	ActionName	VulnerableApp.Controllers.AuthController.Login (VulnerableApp)
✓ X	ConnectionId	0HNMONCBKIK49
✓ X	IP	::1
✓ X	MachineName	DESKTOP-H4CTBTI
✓ X	RequestId	0HNMONCBKIK49:00000002
✓ X	RequestPath	/auth/login
✓ X	SourceContext	VulnerableApp.Controllers.AuthController
✓ X	Username	user1

Validar que no se guarden contraseñas

02 Jul 2026 22:27:50.853

Inicio AuthController.Login (POST) | Intento de usuario: user1 | IP: ::1 | Password: [REDACTED]

Event Level (Information) Type (0x95DD4218) Trace (0411...) Export

✓ X	ActionId	f5b029f9-273d-4031-960b-ae0ff21d0995
✓ X	ActionName	VulnerableApp.Controllers.AuthController.Login (VulnerableApp)
✓ X	ConnectionId	0HNMONCBKIK40
✓ X	IP	::1
✓ X	MachineName	DESKTOP-H4CTBTI
✓ X	RequestId	0HNMONCBKIK40:00000001
✓ X	RequestPath	/auth/login
✓ X	SourceContext	VulnerableApp.Controllers.AuthController
✓ X	Username	user1

- Generar Error.

02 Jul 2026 22:12:33.129

● An unhandled exception has occurred while executing the request.

Event Level (Error) Type (0xBD577466) Trace (76fd...) Export

✓ x	ConnectionId	0HNMON55R7D0Q
✓ x	EventId.Id	1
✓ x	EventId.Name	UnhandledException
✓ x	MachineName	DESKTOP-H4CTBTI
✓ x	RequestId	0HNMON55R7D0Q:00000001
✓ x	RequestPath	/auth/login
✓ x	SourceContext	Microsoft.AspNetCore.Diagnostics.DeveloperExceptionPageMiddleware

```
System.Exception: Error provocado para evidencia en AuthController
at VulnerableApp.Controllers.AuthController.Login() in
C:\Users\valec\OneDrive\Documentos\git\VulnerableApp\Controllers\AuthController.cs:line 32
at lambda_method1(Closure, Object, Object[])
at
Microsoft.AspNetCore.Mvc.Infrastructure.ActionMethodExecutor.SyncActionResultExecutor.Execute(ActionContext
actionContext, IActionResultTypeMapper mapper, ObjectMethodExecutor executor, Object controller, Object[]
arguments)
at Microsoft.AspNetCore.Mvc.Infrastructure.ControllerActionInvoker.InvokeActionMethodAsync()
at Microsoft.AspNetCore.Mvc.Infrastructure.ControllerActionInvoker.Next(State& next, Scope& scope, Object&
state, Boolean& isCompleted)
at Microsoft.AspNetCore.Mvc.Infrastructure.ControllerActionInvoker.InvokeNextActionFilterAsync()
```

CommentController

- Verificar que cada acción genere Information.

02 Jul 2026 22:36:14.924

02 Jul 2026 22:36:14.923

02 Jul 2026 22:36:14.922

02 Jul 2026 22:36:14.913

02 Jul 2026 22:36:14.905

02 Jul 2026 22:36:14.902

02 Jul 2026 22:36:14.896

02 Jul 2026 22:36:14.859

Request starting HTTP/1.1 GET http://localhost:5138/VulnerableApp.styles.css?v=pGi5QHI7edwYw-oFSAhNLPfbjP5k...

Request starting HTTP/1.1 GET http://localhost:5138/css/site.css?v=j6fhJSuuyLpOSLuPU0TsDV0iNjor5S3rDnvxJrt4...

- Endpoint 'null' with route pattern 'lib/bootstrap/dist/css/bootstrap.min.css' is valid for the request path '/lib/bootstrap...
- Endpoint 'null' with route pattern 'lib/bootstrap/dist/css/bootstrap.min.css' is valid for the request path '/lib/bootstrap...
- 3 candidate(s) found for the request path '/lib/bootstrap/dist/css/bootstrap.min.css'

Request starting HTTP/1.1 GET http://localhost:5138/lib/bootstrap/dist/css/bootstrap.min.css - null null

Request finished HTTP/1.1 GET http://localhost:5138/Comment - 200 null text/html; charset=utf-8 167.6834ms

Event Level (Information) Type (0xDDFF70E3) Trace (ee73...) Export

✓ x	ConnectionId	0HNMONCBKIK4L
✓ x	ContentLength	
✓ x	ContentType	text/html; charset=utf-8
✓ x	ElapsedMilliseconds	167.6834
✓ x	EventId.Id	2
✓ x	Host	localhost:5138
✓ x	MachineName	DESKTOP-H4CTBTI
✓ x	Method	GET
✓ x	Path	/Comment
✓ x	PathBase	
✓ x	Protocol	HTTP/1.1
✓ x	QueryString	
✓ x	RequestId	0HNMONCBKIK4L:00000004
✓ x	RequestPath	/Comment
✓ x	Scheme	http
✓ x	SourceContext	Microsoft.AspNetCore.Hosting.Diagnostics

- Generar Warning

02 Jul 2026 22:36:09.132

● El usuario user1 intentó enviar un comentario vacío.

Event Level (Warning) Type (0x8A983B0D) Trace (4fab...) Export

✓ x	ActionId	e618f8e0-c126-4dbe-bca5-8eec377c764e
✓ x	ActionName	VulnerableApp.Controllers.CommentController.AddComment (VulnerableApp)
✓ x	ConnectionId	0HNMONCBKIK4K
✓ x	MachineName	DESKTOP-H4CTBTI
✓ x	RequestId	0HNMONCBKIK4K:00000002
✓ x	RequestPath	/Comment/AddComment
✓ x	SourceContext	VulnerableApp.Controllers.CommentController
✓ x	Usuario	user1

- Generar Error.

02 Jul 2026 22:18:08.491	● Error en CommentController.Index
Event	Level (Error) Type (0x0EBB33FB) Trace (c417...) Export
✓ x	ActionId 87c35cf0-d2d6-4406-9964-781f65a9ec5b
✓ x	ActionName VulnerableApp.Controllers.CommentController.Index (VulnerableApp)
✓ x	ConnectionId 0HNMON8ATFKG3
✓ x	MachineName DESKTOP-H4CTBTI
✓ x	RequestId 0HNMON8ATFKG3:00000002
✓ x	RequestPath /Comment
✓ x	SourceContext VulnerableApp.Controllers.CommentController
System.Exception: Error provocado para evidencia en CommentController at VulnerableApp.Controllers.CommentController.Index() in C:\Users\valec\OneDrive\Documents\git\VulnerableApp\Controllers\CommentController.cs:line 29 at lambda_method3(Closure, Object, Object[])	

ApiController

- Verificar que cada acción genere Information.

02 Jul 2026 22:37:46.757	Executed endpoint 'VulnerableApp.Controllers.ApiController.GetAllUsers (VulnerableApp)'
02 Jul 2026 22:37:46.754	Executed action VulnerableApp.Controllers.ApiController.GetAllUsers (VulnerableApp) in 225.5526ms
02 Jul 2026 22:37:46.690	Executing OkObjectResult, writing value of type 'System.Collections.Generic.List`1[<>f__AnonymousType0`3[System.I...
02 Jul 2026 22:37:46.686	● Selected output formatter 'Microsoft.AspNetCore.Mvc.Formatters.SystemTextJsonOutputFormatter' and content type ...
02 Jul 2026 22:37:46.682	● Attempting to select the first formatter in the output formatters list which can write the result.
02 Jul 2026 22:37:46.679	● Attempting to select an output formatter without using a content type as no explicit content types were specified for t...
02 Jul 2026 22:37:46.678	● No information found on request to perform content negotiation.
02 Jul 2026 22:37:46.669	● List of registered output formatters, in the following order: [Microsoft.AspNetCore.Mvc.Formatters.HttpNoContentOut...
02 Jul 2026 22:37:46.664	Fin ApiController.GetAllUsers Tiempo de ejecución: 123 ms
02 Jul 2026 22:37:46.661	● Closed connection to database 'VulnerableDb' on server '(localdb)\mssqllocaldb' (4ms).
02 Jul 2026 22:37:46.656	● Closing connection to database 'VulnerableDb' on server '(localdb)\mssqllocaldb'.
02 Jul 2026 22:37:46.653	● A data reader for 'VulnerableDb' on server '(localdb)\mssqllocaldb' is being disposed after spending 3ms reading resul...
02 Jul 2026 22:37:46.650	● Closing data reader to 'VulnerableDb' on server '(localdb)\mssqllocaldb'.
02 Jul 2026 22:37:46.644	Executed DbCommand (13ms) [Parameters=[], CommandType='Text', CommandTimeout='30'] SELECT [u].[Id], [u].[User...
02 Jul 2026 22:37:46.632	● Executing DbCommand [Parameters=[], CommandType='Text', CommandTimeout='30'] SELECT [u].[Id], [u].[Username], ...
02 Jul 2026 22:37:46.630	● Initialized DbCommand for 'ExecutingReader' (6ms)

- Generar Error.

02 Jul 2026 22:22:16.882	Fin ApiController.GetAllUsers Tiempo de ejecución: 14 ms
02 Jul 2026 22:22:16.873	● Error en ApiController.GetAllUsers para el usuario Anónimo
Event	Level (Error) Type (0x8E561A84) Trace (d1b7...) Export
✓ x	ActionId 2eddbce5-8b22-4b10-b3e2-e01656e1c3a3
✓ x	ActionName VulnerableApp.Controllers.ApiController.GetAllUsers (VulnerableApp)
✓ x	ConnectionId 0HNMON8ATFKG5
✓ x	MachineName DESKTOP-H4CTBTI
✓ x	RequestId 0HNMON8ATFKG5:00000003
✓ x	RequestPath /api/users
✓ x	SourceContext VulnerableApp.Controllers.ApiController
✓ x	Usuario Anónimo
System.Exception: Error provocado para evidencia en CommentController at VulnerableApp.Controllers.ApiController.GetAllUsers() in C:\Users\valec\OneDrive\Documents\git\VulnerableApp\Controllers\ApiController.cs:line 75	

- Generar Warning

02 Jul 2026 22:37:56.225	● Intento de acceso no autorizado a la API GetAllUsers desde IP: ::1
Event	Level (Warning) Type (0x169D543D) Trace (a057...) Export
✓ x	ActionId 3257bcb6-e216-4cf7-a128-7e7fb0977bae
✓ x	ActionName VulnerableApp.Controllers.ApiController.GetAllUsers (VulnerableApp)
✓ x	ConnectionId 0HNMONCBKIK4N
✓ x	IP ::1
✓ x	MachineName DESKTOP-H4CTBTI
✓ x	RequestId 0HNMONCBKIK4N:00000001
✓ x	RequestPath /api/users
✓ x	SourceContext VulnerableApp.Controllers.ApiController

Hallazgos

Tras la implementación de Serilog y Seq, y la instrumentación detallada de los controladores en VulnerableApp, se identificaron los siguientes hallazgos:

1. Es posible visualizar en tiempo real (mediante nivel Warning) los intentos repetidos de inicio de sesión fallidos
2. La instrumentación en el SearchController reveló la importancia de monitorear la longitud y el contenido de los parámetros mediante el nivel warning cuando un usuario introduce cadenas sospechosamente largas.
3. Se comprobó que el uso del identificador de sesión y la IP en conjunto con el tiempo de ejecución, es posible conocer el comportamiento exacto de un usuario a través de la plataforma.
4. Cualquier caída de la base de datos o error lógico se captura inmediatamente como un evento de nivel Error